

Supplementary Material: Cross-Age Face Verification from Mined Same-Person Social-Media Pairs

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This supplement collects material that supports, but need not occupy, the main paper: (SI) an explicit claims-vs-evidence map; (SII) a consolidated audit of the automatic (LLM / clustering / dedup) supervision components plus a small manual validation (100 posts), with a large multi-annotator audit left beyond this study’s resources; and (SIII) a model / data card. Section and table references of the form “Sec. V” or “Table X” point to the main paper.

I. CLAIMS VS. EVIDENCE

Table I maps every substantive claim to its evidence and its scope, separating what is *established within the attribution protocol* from what is *not established*. The intent is to make the bounded reading explicit and to forestall over-generalization.

II. CONSOLIDATED AUDIT OF THE AUTOMATIC SUPERVISION

The supervision is *naturally* (not human-) supervised, so its automatic components are potential hidden label noise. Table II consolidates the cross-checks reported across the main paper into a single view, in the structure a reviewer would expect, and states explicitly which rows are *automatic cross-checks* (available at corpus scale) and which are checked against *human ground truth*. We additionally ran a *small* manual validation (one annotator; 100 posts and sampled pairs/merges) that broadly confirms the automatic pipeline (Table II, lower block); a large multi-annotator audit with inter-annotator κ remains labor-intensive and beyond this study’s resources. We do not treat the LLM-vs-regex agreement as accuracy against a human reference; the “Reference” column makes the distinction.

The automatic rows are reproducible from the released code and aggregates. The manual-validation rows are a *small* single-annotator spot-check (no inter-annotator κ): they indicate residual error at the few-percent level and corroborate, but do not certify, the automatic pipeline. A large stratified multi-annotator audit (post $\rightarrow$ group $\rightarrow$ cluster $\rightarrow$ pair, with a confusion matrix and κ) remains labor-intensive and beyond this study’s resources, and the residual noise is acknowledged as a limitation.

III. MODEL / DATA CARD

Model. A deliberately weak trainable face recognizer (FaceNet, Inception-ResNet-v1, casia-webface init) fine-tuned with a pair-contrastive margin objective on mined same-person

pairs; this is an *attribution probe*, not a deployable recognizer. **Intended use.** Research into whether naturally occurring longitudinal social-media pairs supply useful cross-age identity supervision. Permitted applied analogues are limited to rights-cleared, consent-based, human-in-the-loop settings (Sec. VIII). **Out of scope / forbidden.** Mass identification, surveillance, de-anonymization, automated decisions, any processing of minors’ data without a legal basis, and use as a general-purpose verifier (the gain is concentrated on the large-gap subset rather than uniform).

Training data. Public multi-photo “then/now” posts (VK communities; an independent Reddit board for the cross-platform test). Supervision is manual-label-free; ages and group integrity come from an LLM caption parser; persons from embedding clustering. Apparent-attribute skew: 76.7% apparent-female. Raw images are *not* released.

Evaluation data. External: FG-NET (large-gap ≥ 25 y subset), AgeDB-30, CALFW, LFW. Internal: a curated cross-age test (5,107 positives with controlled negatives). Cross-platform: 1,575 Reddit positive pairs.

Metrics. ROC-AUC (primary: FG-NET large-gap), EER, TAR@FAR, FMR/FNMR by stratum, bootstrap and identity-level bootstrap CIs, DeLong and paired-permutation tests.

Ethical considerations. Biometric special-category data; no individual consent; minors present in the child $\leftrightarrow$ adult regime and withheld from release; weights and embeddings are biometric templates shared on request under a brief research-use agreement; the de-identified dataset is available on request, and formal ethics documentation will be provided if required (Sec. VIII).

Caveats. The gain is targeted to the large-gap regime and to weak/medium backbones; external validity beyond the then/now format is not established; the automatic supervision carries residual, not-yet-human-audited noise.

IV. FIGURES REFERENCED FROM THE MAIN PAPER

These five plots duplicate data already tabulated in the main paper (headroom curve, data-scaling law, objective comparison, apparent-attribute fairness audit and the external-benchmark summary); they are placed here so the manuscript stays within the 10-page limit for regular papers. No number appears here that is not in the main paper’s tables.

TABLE I

CLAIMS VS. EVIDENCE. “EST.” = ESTABLISHED WITHIN THE WEAK/MEDIUM-BACKBONE ATTRIBUTION PROTOCOL OF THE MAIN PAPER; “BOUNDED” = HOLDS ONLY UNDER STATED CONDITIONS; “OPEN” = NOT ESTABLISHED / OUT OF SCOPE.

Claim	Evidence (main paper)	Status
Mined “then/now” pairs improve large-gap (≥ 25 y) cross-age verification	FG-NET large-gap ROC-AUC 0.736 $\rightarrow$ 0.848; DeLong $p=1.4 \times 10^{-20}$, Δ AUC 95% CI [+0.088, +0.135]; internal 25+ $p=1.7 \times 10^{-34}$ (Sec. V-A)	Est.
The gain is <i>targeted</i> , not a uniform verification gain	AgeDB-30 -0.008 ($p=4 \times 10^{-6}$), CALFW null ($p=0.95$), LFW TAR@FAR=0.1% 0.814 $\rightarrow$ 0.816 (Sec. V-A, Sec. VI)	Est.
The same-person pairing, not the caption age anchor, drives the gain	Ablation: explicit age anchors add only +0.009 on the 25+ bucket (Sec. V-A)	Est.
Real mined pairs outperform synthetic aging	+real 0.848 vs. FRAN 0.727 and gap-matched FRAN-gm 0.759 (+0.089); native-resolution control (Sec. V-B)	Est.
The effect reproduces across objective / architecture / source	3 loss families, 2 architectures, 2 communities; ArcFace 0.851 $\approx$ contrastive 0.848; sub-center ArcFace 0.851 (Sec. V-C/E)	Est.
A noise-robust objective adds nothing on curated pairs	Sub-center ArcFace ($K=3$) $\equiv$ vanilla ArcFace on FG-NET large-gap (Sec. V-E)	Est.
The mechanism is removal of an age shortcut	Under age-matched negatives frozen falls 0.640 $\rightarrow$ 0.517 ($\approx$ chance); pairs restore 0.838 (Sec. V-D)	Est.
The supervision is data-efficient	$\approx 96\%$ of the gain at 10% of identities (Sec. V-C)	Est.
No apparent-demographic stratum degrades; errors fall everywhere	Stratified AUC (all CIs > 0); FMR/FNMR at 1% FMR: FNMR down in every stratum (Sec. V-G)	Est.
About half of the raw “positives” were near-duplicates	65,897 $\rightarrow$ 31,865 (-52%), stable over cosine 0.93–0.97 (Sec. III, Table XI)	Est.
The gain transfers beyond the originating platform, in <i>both</i> directions	VK $\rightarrow$ Reddit overall ROC-AUC 0.718 $\rightarrow$ 0.749; Reddit $\rightarrow$ VK FG-NET large-gap 0.736 $\rightarrow$ 0.783 (+0.047 from a 220-person source); small, noisy sets (Sec. V-H)	Bounded
The contribution is specific to weak/medium backbones	Strong saturated backbones show null/negative gains (Sec. V-C)	Bounded (limitation)
Platform-agnostic / cross-cultural generalization	one platform family, then/now format; single non-VK test	Open
Drop-in upgrade to a general-purpose verifier	the gain is concentrated on the large-gap subset; easy benchmarks are flat	Open (excluded)
Human-grade label-noise guarantee	automatic cross-checks + a small manual validation (100 posts; SII); no multi-annotator κ	Bounded
Full SOTA-pipeline parity	only the shared margin objective is trained, not method-specific add-ons	Open (by design)
Legal/ethical clearance for biometric processing	research basis stated; formal approval pending (Sec. VIII)	Open (author/IRB)

TABLE II

CONSOLIDATED AUDIT OF THE AUTOMATIC SUPERVISION. “REFERENCE” STATES WHAT THE METRIC IS MEASURED AGAINST: AN AUTOMATIC CROSS-CHECK (UPPER BLOCK, CORPUS SCALE) OR HUMAN GROUND TRUTH FROM A SMALL MANUAL VALIDATION (LOWER BLOCK; 100 POSTS + SAMPLED PAIRS/MERGES, ONE ANNOTATOR). n IS THE AUDITED POPULATION. A LARGE MULTI-ANNOTATOR AUDIT WITH κ REMAINS BEYOND RESOURCES.

Audit target	n	Reference	Result
Age extraction (LLM vs. regex)	33,697 captions	automatic cross-check	72.8% agree; on the 27.2% disagreements the LLM is favored (+10.8% filled, -2.5% missed); Table III
Group integrity (LLM classifier)	33,697 posts	automatic (LLM)	92.8% single-person, 6.0% multi-person, 1.0% unknown, 0.2% meme; Table IV
Multi-person noise detector (within-person gender consistency)	multi-photo groups	automatic proxy	2,919 groups with consistency < 0.6 flagged as probable multi-person
Near-duplicate inflation (within-person dedup)	65,897 raw positives	automatic (embedding)	-52% ($\rightarrow$ 31,865); stable for cosine 0.93–0.97; Table XI
Person-clustering recurrence	27,487 post-groups	automatic (embedding)	21,848 persons ($\approx 20\%$ cross-post recurrence); conservative cosine 0.85 merge
Small manual validation (one annotator; 100 posts + sampled pairs/merges)			
Age extraction vs. human	100 captions	human gold	exact-age 89%, within ± 1 y 95% (regex 71% exact); MAE 0.5 y
Group integrity (single vs. multi)	100 posts	human gold	LLM precision 0.96, recall 0.93
Same-person positive pairs	200 pairs	human gold	precision 0.96 (8/200 different-person)
Cluster merges (cosine ≥ 0.85)	100 merges	human gold	95% correct
Near-duplicate removals	100 removed	human gold	93% genuine duplicates

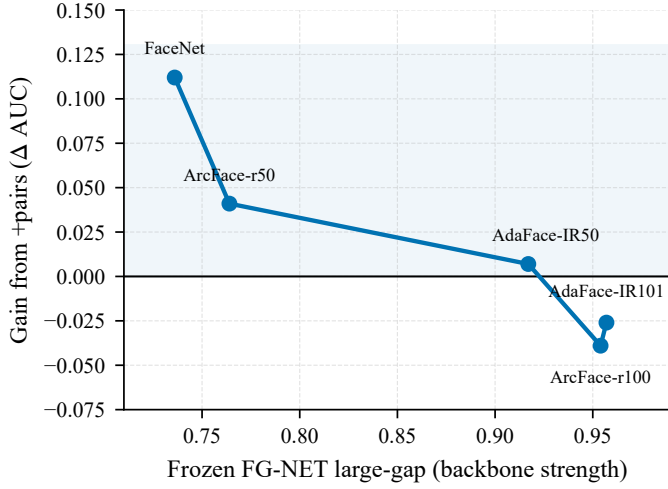


Fig. 1. Headroom curve: the +pairs gain on FG-NET large-gap shrinks monotonically as the frozen backbone strengthens, crossing zero for strong saturated models.

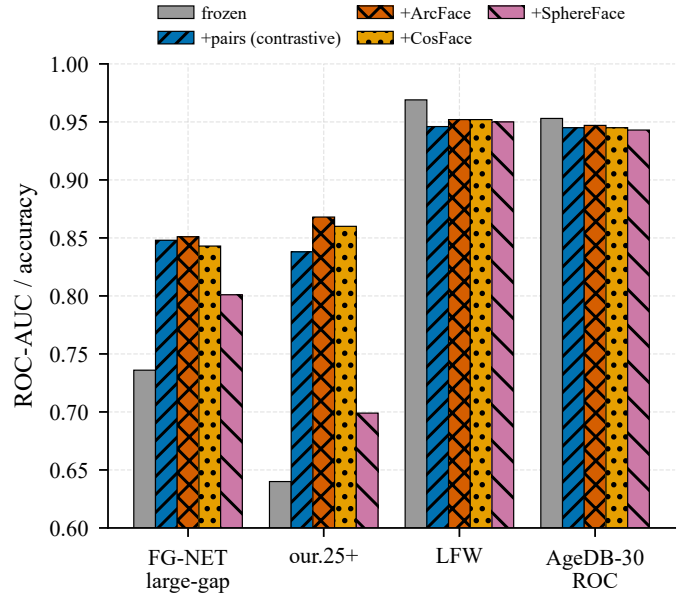


Fig. 3. Objective comparison: contrastive, ArcFace and CosFace coincide on cross-age transfer; SphereFace (hardest to optimize, no λ -annealing) trails. ArcFace/CosFace forget slightly less on easy benchmarks.

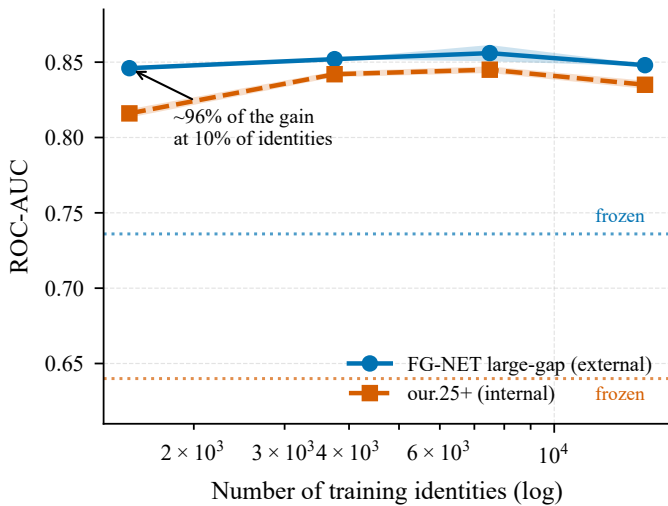


Fig. 2. Data-scaling law: external (FG-NET large-gap) and internal (our.25+) ROC-AUC vs. the number of training identities (log scale). Roughly 96% of the gain is reached at 10% of identities; shaded bands are ± 1 std over three seeds and dashed lines are the frozen baselines.

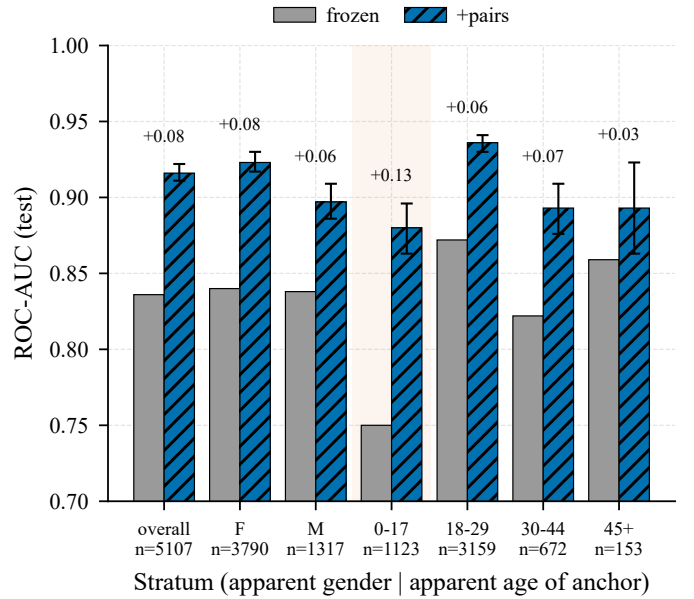


Fig. 4. Apparent-demographic audit: frozen vs. +pairs ROC-AUC across apparent gender/age strata. Every stratum improves; the largest gain is for the youngest (0–17) group, where the frozen baseline is weakest. Error bars are the 95% paired-bootstrap CI of the gain and n is positive pairs per stratum (the sparse 45+ stratum, $n=153$, has a correspondingly wide interval).

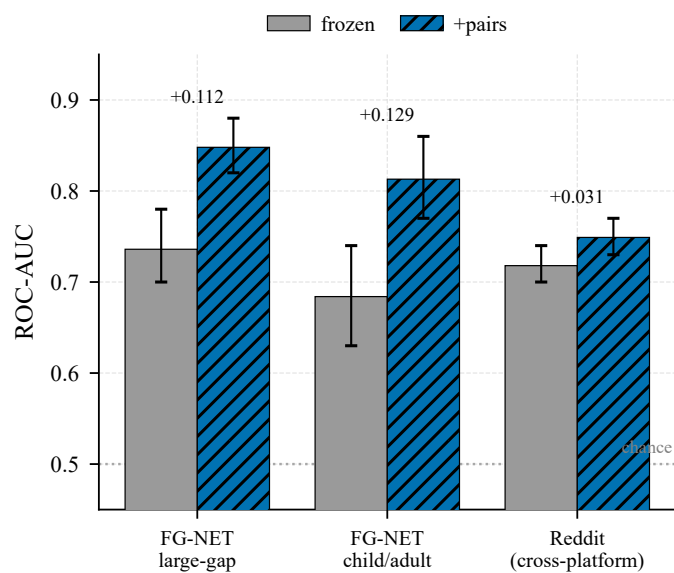


Fig. 5. External generalization. The +pairs gain over the frozen baseline (with 95% CIs) holds across three independent evaluations—FG-NET large-gap, a child↔adult FG-NET subset, and the cross-platform Reddit set—each well above chance. The gain is largest on the in-protocol FG-NET sets and smaller (but positive) on the noisier cross-platform Reddit set.